

# Robotic Hand Exoskeletons for Rehabilitation and Assistance: A State-of-the-Art:

## The core problem

Stroke and similar injuries leave many patients with hand dysfunction that severely limits daily living. Traditional rehab relies on physical therapists, which is labor-intensive, expensive, and inconsistent — therapist fatigue can even cause secondary injury if intensity isn't managed well. Robotic hand exoskeletons offer controllable, repeatable, data-logging alternatives that also seem to boost patient motivation (voluntary participation measurably improves outcomes).

## Methodology

They searched five databases (Web of Science, Scopus, ScienceDirect, IEEE Xplore, PubMed) plus Google Scholar, starting from 851 papers and narrowing to **86** that met inclusion criteria — must describe actual design/control of an *actuated* device (passive orthoses excluded), finger-focused rather than whole-arm devices.

## Hand biomechanics (the design foundation)

A human hand has **21 degrees of freedom**: each of the index/middle/ring/little fingers has 4 DOF, the thumb has 5. Key joints: MCP (2 DOF: flexion/extension + adduction/abduction), PIP and DIP (1 DOF each, flexion/extension only). The paper gives a full range-of-motion table (e.g., MCP flexion 0–90°, PIP 0–100°) that acts as the design ceiling — exoskeletons must never exceed these ROMs, to avoid injury.

## Actuation systems — the heart of the review

Five main approaches, each with trade-offs:

- **Electric motor-driven:** fast, efficient, precise — but motors add weight/inertia directly on the hand.
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- **Serial elastic actuators (SEA):** spring-in-series design gives safe, accurate force control — but adds mechanical complexity.
- **Pneumatic artificial muscles (PAM):** soft, biomimetic, high power-to-weight — but nonlinear/time-varying behavior makes precise control hard.
- **Shape memory alloy (SMA):** very light, low power draw, large force output — but slow response, nonlinearity, and hysteresis limit control precision.

The paper walks through specific real devices for each category with concrete numbers (weights, forces, speeds) — this is the most data-rich section.

### Human-machine kinematic compatibility

Since the hand's joint axes are hard to pin down precisely (soft tissue, individual variability), misalignment between exoskeleton and finger joints creates parasitic forces that can injure the wearer. Three design strategies are compared:

1. **Aligned/overlapping rotation axes** — simple but poor tolerance for individual variation.
2. **Self-aligning joint axes** (passive DOFs that adapt automatically) — better compatibility, but no longer has an exact 1:1 joint correspondence.
3. **Compliant soft gloves** — inherently conform to the hand, most comfortable, but can lack stiffness/force output.

## Degrees-of-freedom design

Trade-off between how many DOFs to actuate (richer motion, more ADL coverage) versus complexity/cost/weight. Most devices prioritize **flexion/extension** since it covers most ADLs, often leaving adduction/abduction passive or ignored. Thumb design gets special attention since thumb loss  $\approx$  40% loss of hand function, and it's involved in over half of grasping movements.

## Control strategies

1. **Position control** — good for early-stage rehab (passive, guided movement).
2. **Force/torque control** — enables safe physical interaction.
3. **Assist-as-needed control** (impedance/admittance-based) — adapts assistance level to what the patient can do themselves, boosting active participation.
4. **Bio-electric signal control** (EMG/EEG/EOG) — uses the wearer's own muscle/brain signals to predict intent; includes clever "bilateral" approaches where a healthy-side EMG signal drives the exoskeleton on the affected hand.
5. Hybrid/combined strategies (e.g., switching passive↔active automatically, master-slave force reproduction).

## Summary table (Table 2)

The paper includes a comparison table of ~17 specific devices with weight, actuation type, DOFs, control strategy, materials, and fingertip force — genuinely useful if you want to compare concrete implementations.

## Discussion & future directions

- No actuation type is universally best — selection depends on patient need, cost, and control complexity.
- Bio-electric control is highlighted as especially promising because it's non-invasive and reflects real motor intent, though EMG/EEG both have signal-quality limitations (fatigue, low spatial resolution, individual variability).
- Future directions flagged: new actuators (piezoelectric, electroactive polymers, MR-SEAs), misalignment-compensation mechanisms, 4D printing with smart materials, and hybrid EMG+EEG control.

## Learning to suppress tremors: a deep reinforcement learning-enabled soft exoskeleton for Parkinson's patients:

Endrei, Földi, Makk & Cserey (2025), *Frontiers in Robotics and AI*. This is quite a departure from the previous two papers — it's a **simulation and control-algorithm** study, not a hardware/experimental validation study. No physical device is tested on a human here; everything happens in physics simulation.

### The core idea

Rather than building a new exoskeleton, this paper builds a **learning-based controller** — specifically a deep reinforcement learning (RL) agent — that decides how much force a soft shoulder/elbow exoskeleton (from earlier work by the same group) should apply to cancel out Parkinsonian tremor, *without* disrupting the patient's intended voluntary movement.

### Motivation — what's wrong with existing approaches

- Traditional exoskeleton control (repetitive control, PID, frequency filtering) has mostly been validated on **static postures** and **healthy people mimicking tremor**, not real dynamic movement or genuine multi-harmonic Parkinsonian tremor.
- These methods usually don't account for **tremor propagation** — how a tremor generated at one joint mechanically couples into torques at neighboring joints (a shoulder tremor bleeding into elbow torque, etc.) — and they struggle to scale to dynamic, everyday movement without manual rule-tuning per activity.
- DRL is proposed as an alternative because it can be trained entirely in simulation (safe, cheap, no patient burden) and can, in principle, generalize across movement types.

### Methods — three main pieces

## 1. Physical simulation of arm + tremor + exoskeleton

- Built in PyBullet + OpenAI Gym, modeling a torso + right arm wearing a 7-actuator soft exoskeleton (shoulder and elbow).
- **Reference (voluntary) movements:** four real motion patterns — shoulder flexion/extension, shoulder abduction/adduction, shoulder external rotation, elbow flexion/extension — captured via IMU sensors on a real subject and converted to joint angles via an extended Kalman filter.
- **Simulated tremor:** modeled as a Parkinson's-realistic process — superposition of two sine waves (two frequency harmonics) capturing ~96.4% of real tremor energy, with randomized amplitude/frequency and Gaussian noise, based on established tremor literature.
- **Joint dynamics:** a full 7-DOF (shoulder + elbow + wrist axes) coupled inertia/damping/stiffness model, so tremor **propagation** between anatomically coupled joints is captured — this is one of the paper's distinguishing features versus prior single-joint studies.
- **Dynamics randomization:** at the start of each training episode, anatomical parameters, actuator precision, actuator slippage, and tremor frequency/amplitude are randomly sampled from set ranges (Table 1) to build robustness against the sim-to-real gap.

## 2. The RL controller

- Uses **TD7**, a recent actor-critic RL algorithm (an evolution of TD3) with an encoder network that learns useful state embeddings, prioritized experience replay, and policy checkpointing — chosen over PPO/TD3/SAC for its sample efficiency.
- **Observation:** 80-dimensional vector of recent actuator forces, tremor torque, actuator positions, and joint positions.
- **Action:** 7 continuous force outputs (one per actuator).
- **Reward function** — a weighted sum of five terms:
  1. Encourage suppression across all tremor-affected axes
  2. Reduce the actual tremor torque
  3. Minimize actuator force used (efficiency/safety)
  4. Smooth actuator force changes (avoid jerky actuation)
  5. Avoid creating unwanted torque on axes that *aren't* trembling (don't disturb voluntary movement)
- Training-specific tweaks: replay buffer split by reference-movement type (to handle uneven trajectory lengths), reduced policy noise, and **pink noise** (rather than white noise) for exploration — found necessary because small actuator-force changes can produce disproportionately large skeletal torque effects.

## Results

- Evaluated over 100 episodes per movement type, using out-of-training-distribution randomization to test generalization.

- The **same trained controller** generalized across all four movement types and nearly all combinations of tremor-affected joint axes (all but one tremor-axis combination were successfully suppressed).
- Tremor amplitude suppression: **up to ~99–100%** in peak cases; median suppression ranged roughly **68–88%** depending on the specific movement (best: shoulder internal/external rotation, median 88.4%; worst: elbow flexion/extension, median 67.9%).
- Qualitatively, the suppressed joint trajectory tracked much closer to the original voluntary trajectory than the tremor-corrupted one — and torque plots showed the controller correctly applying near-zero force on non-tremor axes (respecting the "don't interfere with voluntary movement" goal) while actively countering torque on the tremor-affected axis.

### Limitations (authors' own admissions — notably candid)

- **This is simulation-only** — no physical hardware or patient testing yet. Domain/dynamics randomization is the stand-in for real-world validation, not a substitute for it.
- Performance is inconsistent across movement types and tremor combinations (shoulder flexion/extension performed worst), suggesting the actuator layout itself may need rethinking, possibly combined with FES.
- Control runs on a dense neural-network actor, which can produce large frame-to-frame force swings that a traditional PID stage can't easily smooth — flagged as a real hardware deployment risk given typical actuator control frequencies (30–40 Hz).
- Relies on full (Markov) state observability — real sensors are noisier and partially observable, so future work would need state estimation (e.g., Kalman filtering) layered in.
- Only validated on movement trajectories similar to training data; genuinely novel movements may break the controller.
- No safety layer (torque/acceleration limits) yet — flagged as essential before real deployment.

### How this fits with the other two papers you've reviewed

This is a nice contrast piece:

- The **Chen et al. review** covers hand exoskeleton *hardware* design broadly (actuators, DOFs, kinematic compatibility) and *classical* control strategies (position, force, admittance, EMG-based).
- The **HELM paper** is a concrete hardware instance using simple open-loop position control for **passive** tremor suppression in the *hand*.
- This **Endrei et al. paper** is almost the opposite end of the spectrum: no new hardware, instead a sophisticated **learned, closed-loop, active-mode** controller for the **shoulder/elbow**, explicitly trying to solve the dynamic-movement and personalization problems that traditional control (as used in HELM-style devices) can't handle. It's essentially arguing "instead of hand-designed PID/position control, let an RL agent learn

to suppress tremor while preserving intentional motion" — a control philosophy that could eventually be applied to hand exoskeletons like HELM if extended to active/dynamic scenarios.

## A Hand Exoskeleton with Linear Motors (HELM) for Pathological Tremor Suppression of Fingers

Yi, Zhang, Lau & Zhang (2025), *Intelligent Sports and Health*. This is a design + experimental study (not a review) introducing what the authors claim is the **first hand exoskeleton to suppress tremor across all five fingers**.

### The problem

Pathological tremor (involuntary, rhythmic shaking) affects millions worldwide — roughly 1 million new idiopathic tremor diagnoses/year in the US, and ~1.7% of over-65s in China have Parkinson's, most with tremor. Existing treatments fall short:

- **Drugs:** up to 30% of patients don't respond or can't tolerate side effects; up to 56% eventually quit.
- **Deep brain stimulation (DBS):** effective but invasive, carries hemorrhage/psychiatric risks, and only 1.6–4.5% of PD patients are even eligible.

This leaves **wearable tech** (functional electrical stimulation or robotic exoskeletons) as a management (not cure) option. FES has muscle-fatigue and poor-selectivity issues. Existing tremor-suppression exoskeletons almost all target the **wrist or elbow** — the authors note this is the first device targeting the **fingers** specifically, and the first with >4 DOF (most prior systems: 1–3 DOF).

### Device design (HELM)

- **Actuation:** five lightweight linear motors (Firgelli L12), one per finger, each independently controlling flexion/extension — enabling asynchronous finger/gesture movement.
- **Two modes:**
  - **Active mode:** motor actively drives finger motion.
  - **Passive mode:** motor holds a fixed position; the core trick for tremor suppression — tremor is treated as a "disturbance" that the motor's back-driving force resists and damps out.
- **Weight:** exoskeleton body 213 g; full system (battery + control circuit) 402 g.

- **Mechanism:** multilink transmission (D-H kinematic modeling, Jacobian, Lagrangian dynamics all derived), achieving ROM of  $\sim 50^\circ$  at MCP and  $\sim 90^\circ$  at PIP. 3D-printed (SLA resin), donned in under a minute.
- **Control:** STM32 microcontroller, PWM-driven, **open-loop only** (a notable limitation flagged by the authors themselves) — position feedback via ADC to a PC for monitoring only, not for closed-loop correction.

## Experiments — two parts

### 1. Active mode (functional performance)

- Fingertip force: 11–15 N depending on finger.
- Grip force: up to 30 N.
- Grasping test: successfully grasped a water bottle, pen bag, and mango — both in a healthy subject and (qualitatively) in a Parkinson's patient.
- Trajectory tracking (step/sinusoidal/triangular inputs): motor tracked reference trajectories accurately, though errors grew near peak/reversal points (attributed to motor nonlinearity when reversing direction).

### 2. Passive mode (tremor suppression) — the key result

- Tested on **2 healthy subjects** (simulating tremor voluntarily, verified via EMG to ensure consistent tremor amplitude before/after) and **1 real Parkinson's patient** (65F, 6-year PD history, left-hand tremor).
- Used Vicon motion capture (12 markers) + EMG to quantify tremor angle reduction (via % decline in mean absolute value and RMS of finger angle).
- **Results:**
  - Subject 1 (healthy): 86.3% average MAV reduction, 83.7% RMS reduction.
  - Subject 2 (healthy): 91.5% average MAV reduction, 91.1% RMS reduction.
  - **Patient:** 73.4% average MAV reduction, 73.3% RMS reduction.
- Performance was slightly worse on the patient — attributed to her much smaller native tremor angle (less "signal" to suppress against fixed assembly gaps) plus individual hand-fit differences.

### Important scope limitation (tremor types)

The authors carefully distinguish three tremor types: **postural** (holding a position against gravity), **rest** (while relaxed), and **kinetic** (during voluntary movement). Passive mode locks the finger in place — so it suppresses postural/rest tremor well, but **cannot** be used for kinetic tremor, since it would also block the wearer's intended voluntary movement. This study only tested postural tremor suppression; kinetic tremor

suppression is left as future work requiring active-mode closed-loop control with motion-intention recognition (e.g., via EMG).

#### **Stated limitations (authors' own discussion)**

- Open-loop control is inadequate for kinetic tremor suppression (future work needs closed-loop trajectory tracking).
- DIP joint isn't constrained by the exoskeleton, so fingertip tremor isn't fully suppressed.
- Mechanical assembly gaps reduce suppression fidelity, especially for smaller tremor amplitudes (as seen in the patient data).
- Rigid-body design has wearability trade-offs; soft/glove-based alternatives are flagged as a future direction.
- Only 3 subjects tested — small sample, future work needs more patients.

#### **How this connects to the review article you read earlier**

This HELM paper is a nice concrete complement to the Chen et al. review: it's an **electric-motor-driven** (specifically linear-motor), rigid, open-loop **position-controlled** device — one specific point in the exact design space (actuator type, DOF, control strategy) that the review surveyed abstractly. It also opens up an application area (tremor suppression) that sits slightly outside the review's main focus (rehab/ADL assistance after stroke), though the review's actuation and DOF-design frameworks map directly onto it.